Intelligent Customer Churn Prediction Using Data Analytics and Classification Models

PROJECT REVIEW 1 — LITERATURE SURVEY & RESEARCH GAPS

Allipuram Akhil (2420030540) | D. Surya (2420030440)
Faculty Guide: Dr. K. Swapnika
Data Analytics and Visualization (24DSB3101)

Literature window: 2021–2026 | 20 selected research papers

PROJECT DEFINITION

Project Title & Problem Statement

A churn-risk classification study designed to connect predictive performance with usable customer-level insight.

PROJECT TITLE

Intelligent Customer Churn Prediction

Using Data Analytics and Classification Models

Study focus

Predict churn, compare classification models, interpret customer risk and present findings through an analytics dashboard.

PROBLEM STATEMENT

Telecommunication and subscription businesses need to identify customers who may discontinue their services. Churn behaviour is influenced by multiple customer and service attributes, so a single indicator may not be sufficient to distinguish customers at risk. A useful churn-prediction workflow must prepare the data carefully, account for class imbalance, compare classifiers using consistent evaluation measures, and explain why a customer is assigned a particular risk level. This project addresses that need by analysing customer churn data, training and comparing classification models, evaluating both class predictions and ranking performance, and organizing the outputs into an interactive dashboard for individual and batch prediction.

5,000	13	27.3%	22
records	raw columns	churners	encoded features

Dataset and preprocessing figures are reported in the uploaded project-review PPT.

PROJECT DIRECTION

Project Objectives

The objectives connect data preparation, comparative classification, interpretation and an actionable output.

01 Understand the data

Inspect missing values, duplicates, class balance and feature distributions before model development.

02 Prepare predictive features

Apply suitable missing-value treatment, categorical encoding and feature engineering.

03 Compare classifiers fairly

Evaluate Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost and CatBoost using a consistent protocol.

04 Interpret model behaviour

Identify influential customer attributes and explain individual churn predictions.

05 Generate useful predictions

Produce churn probabilities and customer-level risk categories for single and batch inputs.

06 Deliver an interactive dashboard

Bring EDA, model evaluation, prediction and interpretation into one user-facing application.

Research extension: test a hybrid or ensemble candidate only after its model definition and results can be reproduced.

RELATED WORK · 2021–2026

Literature Survey (1/5)

Recent studies on churn modelling, ensemble learning, class imbalance, explainability and decision-support applications.

NO. / YEAR	PAPER TITLE & SOURCE	METHOD / RESEARCH APPROACH	KEY CONTRIBUTION & PROJECT RELEVANCE
01 2021	**Hybrid approach using machine learning algorithms for customers' churn prediction in the telecommunications industry** *Boateng & Fokone (2021) · Concurrency and Computation: Practice and Experience*	Two-layer flexible voting ensemble, evaluated on IBM and Duke datasets under balanced and imbalanced settings.	Motivates combining diverse classifiers rather than relying on a single model. DOI 10.1002/cpe.6627
02 2021 / 2022	**Distributed model for customer churn prediction using convolutional neural network** *Author not specified in source slide · Journal of Modelling in Management (online 2021; issue 2022)*	2-D CNN with preprocessing and Apache Spark distributed processing on Telco Customer Churn data.	Shows a deep-learning and distributed-processing route; accuracy is dataset-specific. DOI 10.1108/JM2-01-2021-0032
03 2022	**Detecting the Risk of Customer Churn in Telecom Sector: A Comparative Study** *Edwine et al. (2022) · Mathematical Problems in Engineering*	Mutual-information feature selection, controlled-ratio undersampling, RF/SVM/KNN and hyperparameter search.	Addresses class imbalance and tuning; supports evaluation beyond accuracy. DOI 10.1155/2022/8534739
04 2023	**Research on customer churn prediction and model interpretability analysis** *Peng, Peng & Li (2023) · PLOS ONE*	Banking churn study combining predictive modelling with interpretability analysis.	Connects churn prediction with explanations for decision-making. DOI 10.1371/journal.pone.0289724

Source note: paper metadata and reported figures are reproduced from the uploaded literature-survey PPT. DOI links are clickable for verification.

RELATED WORK · 2021–2026

Literature Survey (2/5)

Recent studies on churn modelling, ensemble learning, class imbalance, explainability and decision-support applications.

NO. / YEAR	PAPER TITLE & SOURCE	METHOD / RESEARCH APPROACH	KEY CONTRIBUTION & PROJECT RELEVANCE
05 2023	**Early warning of telecom enterprise customer churn based on ensemble learning** *Zhou et al. (2023) · PLOS ONE*	Ensemble-learning approach focused on early warning for enterprise telecom customers.	Frames churn modelling as an early-retention signal rather than only post-churn reporting. DOI 10.1371/journal.pone.0292466
06 2023	**Customer churning analysis using machine learning algorithms** *Authors not specified in source slide · International Journal of Intelligent Networks*	Compared stochastic gradient boosting, Random Forest, KNN and Logistic Regression.	Provides a baseline comparison across linear, neighbour-based and ensemble methods. DOI 10.1016/j.ijin.2023.05.005
07 2023	**Machine learning based customer churn prediction in home appliance rental business** *Suh (2023) · Journal of Big Data*	Used real operating data from a Korean water-purifier rental company; evaluated F1 and AUC.	Shows transfer of churn methods beyond telecom; reported F1 0.93 and AUC 0.88 on that dataset. DOI 10.1186/s40537-023-00721-8
08 2024	**Customer churn prediction in telecom sector using machine learning techniques** *Wagh et al. (2024) · Results in Control and Optimization*	Applied machine-learning techniques to telecom churn data and examined factors associated with leaving.	Supports domain-specific model comparison and analysis of churn-related behaviour. DOI 10.1016/j.rico.2023.100342

Source note: paper metadata and reported figures are reproduced from the uploaded literature-survey PPT. DOI links are clickable for verification.

RELATED WORK · 2021–2026

Literature Survey (3/5)

Recent studies on churn modelling, ensemble learning, class imbalance, explainability and decision-support applications.

NO. / YEAR	PAPER TITLE & SOURCE	METHOD / RESEARCH APPROACH	KEY CONTRIBUTION & PROJECT RELEVANCE
09 · 2024	**Hybrid black-box classification for customer churn prediction with segmented interpretability analysis** *Authors not specified in source slide · Decision Support Systems*	Hybrid segmented models using black-box classifiers, SHAP-based visualisation and benchmark experiments.	Connects predictive performance with segment-level interpretation for retention decisions. DOI 10.1016/j.dss.2024.114217
10 · 2024	**Explaining customer churn prediction in telecom industry using tabular machine learning models** *Authors not specified in source slide · Machine Learning with Applications*	Compared tabular classifiers and applied global/local explanations, including SHAP and visual plots.	Gradient Boosting reported 81% accuracy in that study; demonstrates explanation alongside performance. DOI 10.1016/j.mlwa.2024.100567
11 · 2024	**A Review on Machine Learning Methods for Customer Churn Prediction and Recommendations for Business Practitioners** *Ahmed et al. (2024) · IEEE Access*	Reviewed 212 published articles from 2015–2023 across churn workflow stages and industries.	Identifies profit-oriented metrics as under-researched and recommends ensembles and explainability. DOI 10.1109/ACCESS.2024.3402092
12 · 2024	**A Decade of Churn Prediction Techniques in the TelCo Domain: A Survey** *Barsotti et al. (2024) · SN Computer Science*	Surveyed telecom churn algorithms, datasets, evaluation metrics, issues and open challenges.	Maps domain methods and notes heterogeneous datasets complicate direct score comparisons. DOI 10.1007/s42979-024-02722-7

Source note: paper metadata and reported figures are reproduced from the uploaded literature-survey PPT. DOI links are clickable for verification.

RELATED WORK · 2021–2026

Literature Survey (4/5)

Recent studies on churn modelling, ensemble learning, class imbalance, explainability and decision-support applications.

NO. / YEAR	PAPER TITLE & SOURCE	METHOD / RESEARCH APPROACH	KEY CONTRIBUTION & PROJECT RELEVANCE
13 2024	**Investigating customer churn in banking: a machine learning approach and visualization app for data science and management** *Authors not specified in source slide · Data Science and Management*	Compared machine-learning algorithms and built an R Shiny visualisation application for banking churn.	Supports combining model comparison with an interactive analytics dashboard. DOI 10.1016/j.dsm.2023.09.002
14 2024	**Using topological data analysis and machine learning to predict customer churn** *Sagming, Heymann & Visaya (2024) · Journal of Big Data*	Explored topological data analysis alongside machine learning for churn prediction.	Introduces structural data analysis as an alternative feature and representation perspective. DOI 10.1186/s40537-024-01020-6
15 2025	**A data-driven approach with explainable artificial intelligence for customer churn prediction in the telecommunications industry** *Authors not specified in source slide · Results in Engineering*	XAI-Churn TriBoost combines XGBoost, CatBoost and LightGBM using soft voting, with SHAP and LIME.	Reports more than 2 million records and 96.44% accuracy; results are dataset-specific. DOI 10.1016/j.rineng.2025.104629
16 2025	**Customer Churn Prediction: A Systematic Review of Recent Advances, Trends, and Challenges in Machine Learning and Deep Learning** *Lemmi et al. (2025) · Machine Learning and Knowledge Extraction*	PRISMA-guided review: 240 studies mapped, with 61 examined in detailed synthesis (2020–2024).	Highlights class imbalance, concept drift, explainability and business-aligned metrics as challenges. DOI 10.3390/make7030105

Source note: paper metadata and reported figures are reproduced from the uploaded literature-survey PPT. DOI links are clickable for verification.

RELATED WORK · 2021–2026

Literature Survey (5/5)

Recent studies on churn modelling, ensemble learning, class imbalance, explainability and decision-support applications.

NO. / YEAR	PAPER TITLE & SOURCE	METHOD / RESEARCH APPROACH	KEY CONTRIBUTION & PROJECT RELEVANCE
17 2025 / 2026	**A comprehensive survey on customer churn analysis studies** *Shahabikargara et al. (online 2025; issue 2026) · Journal of Information and Telecommunication*	Survey across customer journeys, features, algorithms and telecom, banking and gaming sectors.	Frames cross-industry differences and possible future research directions. DOI 10.1080/24751839.2025.2528440
18 2025	**An adaptive analytics framework for customer retention through integrative feature optimization and ensemble learning** *Authors not specified in source slide · Decision Analytics Journal*	PCA, feature optimisation and weighted-voting XGBoost/LightGBM with SHAP interpretation.	Reports AUC 0.99 and accuracy 0.98 on its e-commerce dataset; cross-dataset comparison is limited. DOI 10.1016/j.dajour.2025.100626
19 2025	**Explainable Customer Churn Prediction in Telecom Using Ensemble Learning and SHAP Analysis** *Rasool, Aljawaheri & Karram (2025) · Mathematical Modelling of Engineering Problems*	Stacking ensemble of Decision Tree, Random Forest, Extra Trees, Gradient Boosting and XGBoost; SHAP analysis.	Directly relevant to comparing a stacked hybrid with baseline classifiers. DOI 10.18280/mmep.121226
20 2026	**Explainable AI-driven customer churn prediction: a multi-model ensemble approach with SHAP-based feature analysis** *Ahmad & El Hajj (2026) · Frontiers in Artificial Intelligence*	Seven models, soft-voting ensemble, SHAP, customer segmentation and threshold optimisation on Telco data.	Reports XGBoost ROC-AUC 0.932 and ensemble ROC-AUC 0.918; links prediction with interpretation and risk profiles. DOI 10.3389/frai.2026.1748799

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SYNTHESIS

What the Recent Literature Shows

The 20 selected studies point to a consistent set of technical and practical themes.

ML

ENSEMBLE METHODS

Random Forest, Gradient Boosting, XGBoost, CatBoost, LightGBM, voting and stacking are recurring approaches.

DL

DEEP LEARNING

CNN, RNN/LSTM and DNN approaches appear when sequential, behavioural or larger-scale data are available.

XAI

EXPLAINABILITY

SHAP and LIME connect model outputs with influential customer attributes and local predictions.

IMB

CLASS IMBALANCE

Class distribution, sampling strategy and threshold choice affect recall and retention workload.

UI

DECISION SUPPORT

Dashboards and visualisation apps connect model evaluation with individual or operational outputs.

VAL

FAIR COMPARISON

Accuracy and AUC vary by dataset, preprocessing, class balance and validation protocol.

Implication: evaluate models on a shared protocol and report both predictive quality and decision usefulness.

GAP IDENTIFICATION

Research Gaps Identified

Each gap is linked to evidence in the reviewed literature and a concrete response in the proposed project.

01	END-TO-END WORKFLOW	Studies often focus on one model, a review, or a single stage of churn analysis.	PROJECT RESPONSE: Connect data upload and validation, EDA, preprocessing, modelling, evaluation and dashboard output.
02	IMBALANCE HANDLING	Class imbalance and resampling recur across comparative studies and systematic reviews.	PROJECT RESPONSE: Document the class distribution and compare suitable class-weighting or resampling strategies inside training folds.
03	FAIR MODEL COMPARISON	Different datasets and validation protocols make published scores difficult to compare directly.	PROJECT RESPONSE: Evaluate selected baselines on the same split and cross-validation procedure.
04	ACTIONABLE RISK RANKING	A churn label alone does not identify which customers should be prioritised.	PROJECT RESPONSE: Convert churn probabilities into clearly defined risk categories and validate decision thresholds.
05	EXPLAINABILITY + OUTPUT	Explainability studies show value, but explanations need to connect to usable prediction outputs.	PROJECT RESPONSE: Pair feature-level and local explanations with individual and batch prediction results.

INNOVATION

Proposed Innovation & Project Contribution

The contribution is an integrated, carefully evaluated churn workflow rather than a claim of a newly invented classifier.

POSITIONING

From a churn score to a usable decision

The project brings data quality, model comparison, risk categorisation and dashboard-based prediction into one workflow.

Research discipline

A hybrid model is tested only after its definition and results can be reproduced.

01	**Unified workflow**	Bring data validation, EDA, preprocessing, model evaluation and dashboard prediction together.
02	**Imbalance-aware evaluation**	Make class balance visible and assess recall, PR-AUC and confusion-matrix behaviour, not accuracy alone.
03	**Consistent model comparison**	Use a shared split and validation procedure so selected classifiers can be compared fairly.
04	**Customer-level risk output**	Translate churn probabilities into risk categories and support single or batch predictions.
05	**Interpretability with context**	Connect influential customer attributes and local explanations to model outputs, without treating them as causal effects.

FEASIBILITY

Project Feasibility & Current Evidence

A qualitative assessment based on the dataset profile and implementation direction reported in the uploaded PPT.

TECHNICAL

FEASIBLE

The supplied report describes 5,000 rows, 13 columns, 150 missing feature values and no duplicate rows. Ten input features were expanded to 22 encoded features. Model evaluation outputs are already reported.

FEASIBLE

OPERATIONAL

WORKFLOW DEFINED

The proposed dashboard covers EDA, model evaluation, single-customer prediction and batch prediction. Operational readiness depends on completing and verifying the end-to-end dashboard flow.

IN PROGRESS

ECONOMIC

COST DETAILS NOT PROVIDED

The source PPT does not state a project budget, hosting cost, hardware requirement or software licensing cost. These items should be confirmed before making a final economic-feasibility claim.

TO CONFIRM

REPORTED MODEL RESULTS

0.744	0.7635	0.5546	0.6630	0.7706
Random Forest	**Gradient Boosting**	**Gradient Boosting**	**Decision Tree**	**Logistic Regression**
Highest accuracy	Highest ROC-AUC	Highest PR-AUC	Highest recall	Selected 5-fold CV ROC-AUC

Interpretation note: different models lead on different metrics. The source PPT reports Logistic Regression as the five-fold CV ROC-AUC selection (0.7706); the selection rule should be stated explicitly.

PROJECT PLAN

Proposed Project Plan

The phases below translate the seven-step workflow in the source PPT into an ordered implementation plan. No calendar durations were specified in the source.

PHASE 01

DATA INPUT & VALIDATION

Load the customer churn CSV; validate file structure, field types and target labels.

DELIVERABLE

Validated dataset

PHASE 02

EDA & PREPARATION

Inspect missing values, churn distribution and relationships; apply imputation, encoding, feature engineering and imbalance strategy.

DELIVERABLE

Prepared feature set

PHASE 03

MODEL TRAINING

Train baseline classifiers using training data only; preserve a common split and cross-validation protocol.

DELIVERABLE

Comparable model runs

PHASE 04

EVALUATION & INTERPRETATION

Compare Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC and confusion matrices; inspect feature importance and local explanations.

DELIVERABLE

Evaluation and explanation report

PHASE 05

DASHBOARD & FINAL REVIEW

Integrate risk categories, single/batch prediction and history; test workflow and document verified outputs.

DELIVERABLE

Review-ready application and report



Plan is proposed from the source workflow; dates, owners and milestone completion status are to be finalised by the team.

PROJECT REVIEW 1

Thank You

Questions & Feedback Welcome

Allipuram Akhil · 2420030540

Devineni Surya · 2420030440

Faculty Guide: Dr. K. Swapnika